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Every Coin Has Two Sides: On the Dual Nature of Generalization in On- Policy Distillation of Large Language Models

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In-domain distribution shift: OPD在同领域内不同难度、不同语言上具有泛化能力

Cross-domain transfer: 训练 prompt 的 domain 并不严格限制被蒸馏进去的 capability。

Multi-teacher interaction: 路由opd会有跷跷板效应，顺序opd会有遗忘问题

Setting1: policy-gradient style OPD

$$\mathcal{L}_{\text{OPD}}^{\text{PG}}(\theta) = -\mathbb{E}_{x,y} \left[\frac{1}{T} \sum_{t=1}^T \bar{A}_t^{\text{OPD}} \log \pi_{\theta}(y_t | h_t) \right], \quad \hat{A}_t^{\text{OPD}} = \text{sg} [\log \pi_{\phi}(y_t | h_t) - \log \pi_{\theta}(y_t | h_t)],$$

强化 teacher 比 student 概率高的sample, 反之抑制

Setting2: Same/Cross-Origin OPD:

Teacher和student来自同一个base model, 则成为same origin (例如sft和rl版本)

Setting3: In/Cross-Domain Generalization:

problem difficulty, language, and reasoning horizon

在一个上面训练, 其他测试: math, code, science, and instruction following (IF)

研究的问题

200 steps x 128 bs

Math: trained on BigMath

English Math (AMC2023, MATH-500, AIME2025, AIME2026, BeyondAIME, OlymMATH-Hard)
as the primary evaluation benchmarks

Chinese Math (OlymMATH-ZH, LiveMathBench-ZH) for the language shift (generalizing from English to Chinese)

Long-Horizon Math (the AIME24-Horizon-2 and AMC23-Horizon-4) generalizing to problems that require more reasoning steps

Code: trained on DeepCoder-Preview-Dataset , evaluate on LiveCodeBench

Science: trained on TextbookReasoning and SCP-116K, evaluate on GPQA-Diamond

IF: train on Nemotron-Post-Training-IF, evaluate on IF-Eval

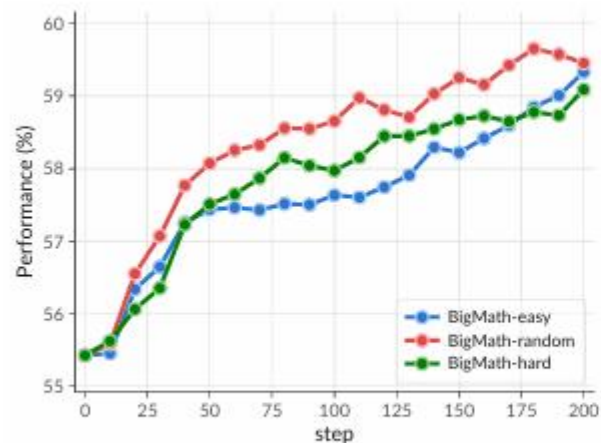
实验1：难度泛化 (teacher侧)

OPD teaches student models the teachers' reasoning patterns rather than the correct answers to particular problems.

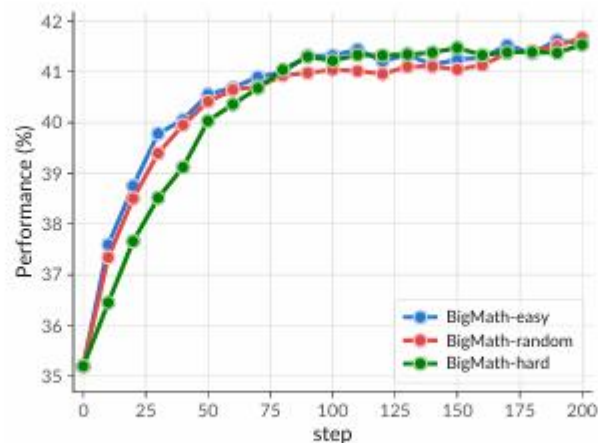
[1] problems the teacher never solves are as useful as those it always solves

(三条曲线收敛到同一个水平)

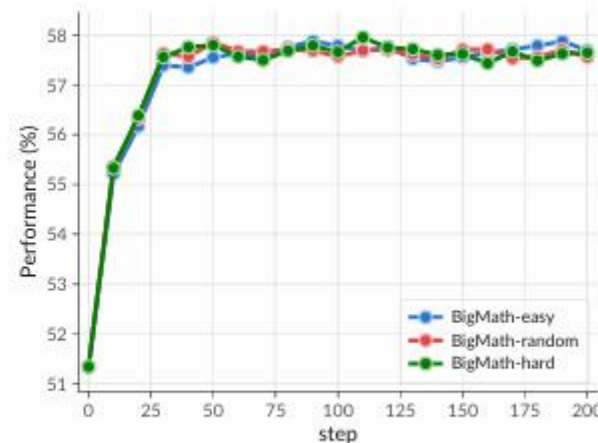
[2] even grade-school problems recover most of the OPD gain.



(a) Qwen3-32B → Qwen3-8B-SFT



(b) Polaris-7B → DS-distill-1.5B



(c) Polaris-7B → DS-distill-7B

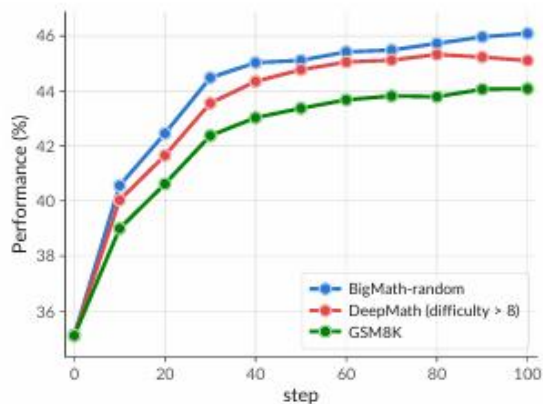
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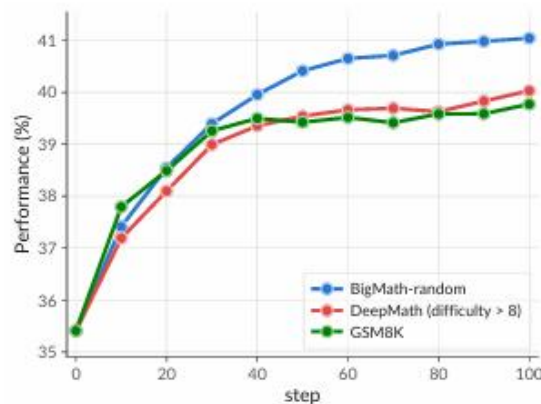
[1] problems the teacher never solves are as useful as those it always solves

[2] even grade-school problems recover most of the OPD gain.

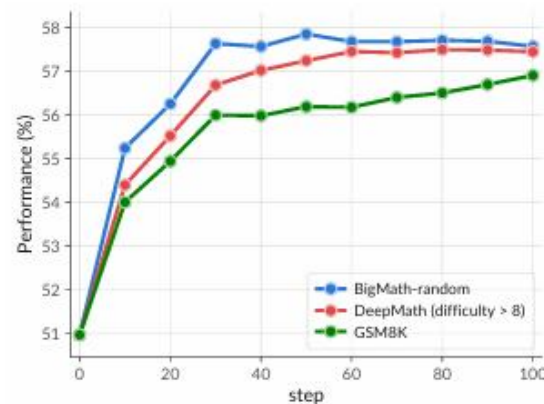
(极度简单和极度困难上学习，结果指标差距<两个点，但是数据多样性有增益)



(a) JustRL-1.5B → DS-distill-1.5B



(b) Polaris-7B → DS-distill-1.5B



(c) Polaris-7B → DS-distill-7B

实验1：难度泛化 (student侧)

student 已经稳定掌握的问题，可以不再持续让 teacher realign

个人认为差距不大

Filtering Strategy	AMC2023	MATH500	AIME2025	AIME2026	BeyondAIME	OlymMATH-Hard	Average
<i>Teacher: Polaris-7B, Student: DS-distill-1.5B</i>							
w/o dynamic sampling	80.4%	89.5%	30.2%	30.1%	13.6%	4.7%	41.4%
pass-rate = 0	78.8%	90.1%	30.5%	30.0%	14.7%	4.4%	41.4%
pass-rate = 1	79.9%	89.7%	30.9%	28.9%	14.7%	4.0%	41.4%
pass-rate $\in [0, 1)$	80.9%	89.8%	31.2%	31.1%	13.7%	5.4%	42.0%(+0.6 pp)
<i>Teacher: Light-R1-14B, Student: DS-distill-7B</i>							
w.o. dynamic sampling	91.7%	95.0%	42.8%	49.7%	26.5%	8.7%	52.4%
pass-rate = 0	92.1%	94.3%	40.2%	50.3%	28.2%	7.8%	52.1%(-0.3 pp)
pass-rate = 1	92.3%	94.5%	42.2%	48.7%	28.0%	8.0%	52.2%(-0.2 pp)
pass-rate $\in [0, 1)$	92.1%	95.4%	43.2%	50.6%	28.0%	7.5%	52.8%(+0.4 pp)

实验 2：语言泛化，horizon泛化

Two students, DS-distill-1.5B/7B, trained on BigMath-random

each paired with a same-origin and a cross-origin teacher

OPD improves: in-domain, Chinese, long-horizon (长期基准上的问题需要长期推理并跨子问题传播中间答案, 是训练数据中不存在的结构)

(1) opd有泛化性

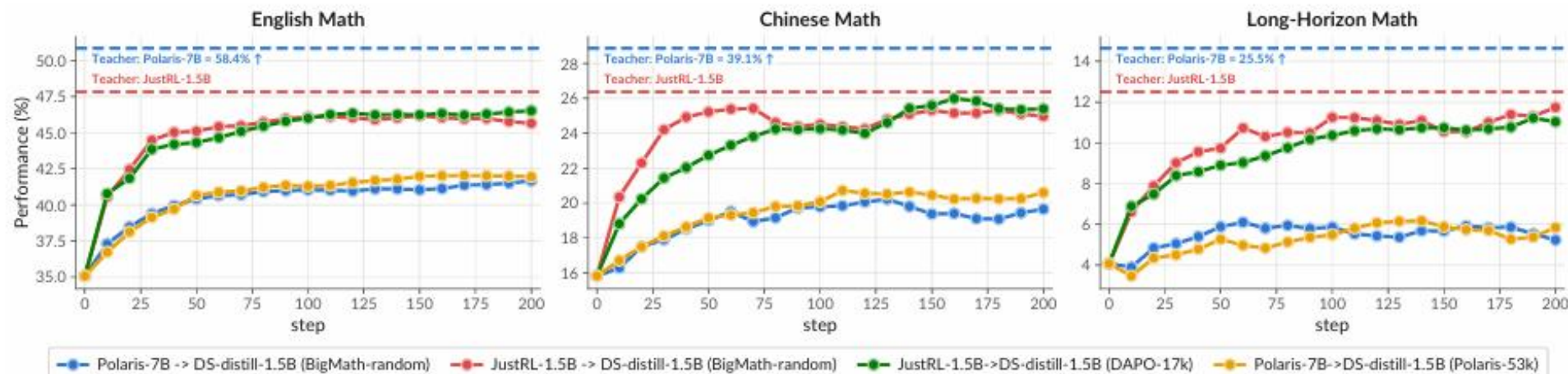
(2) same-origin teacher效果更好, 尽管非同源的teacher更强

(3) teacher 训练时用的数据不会让 OPD 效果更好 (justrl: dapo-15k, polaris: Polaris-53k)

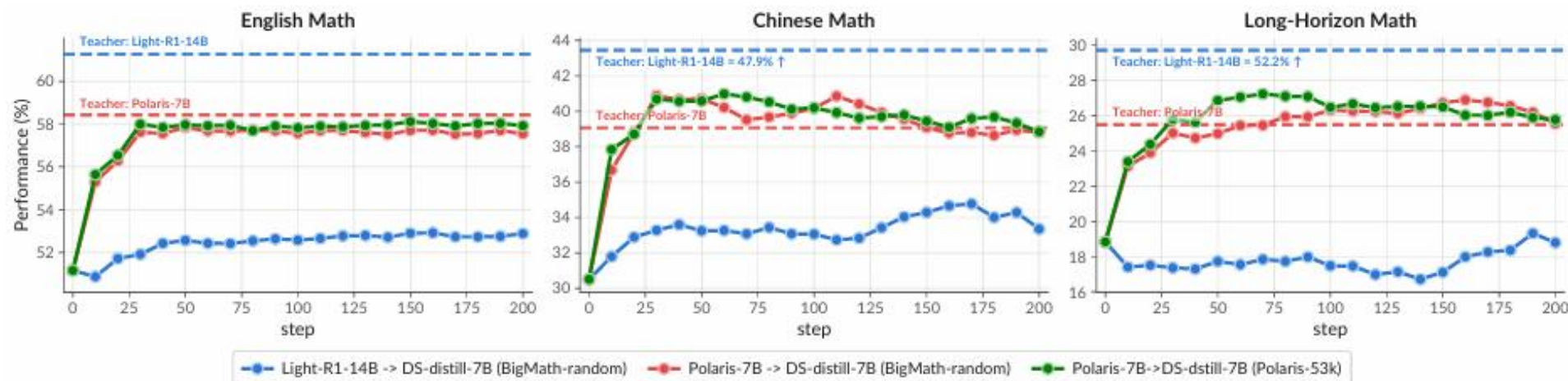
实验 2：语言泛化，horizon泛化



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(a) DS-distill-1.5B

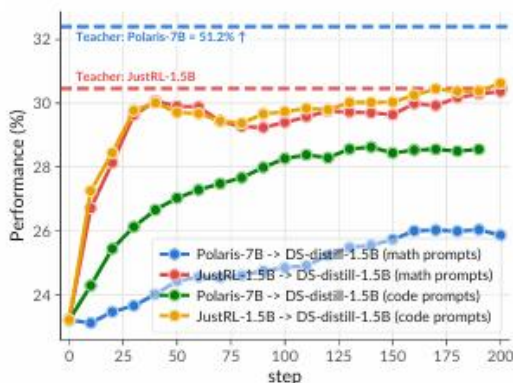


(b) DS-distill-7B

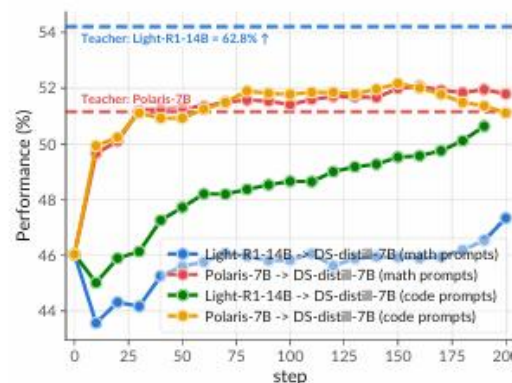
实验 3：领域泛化, math to code



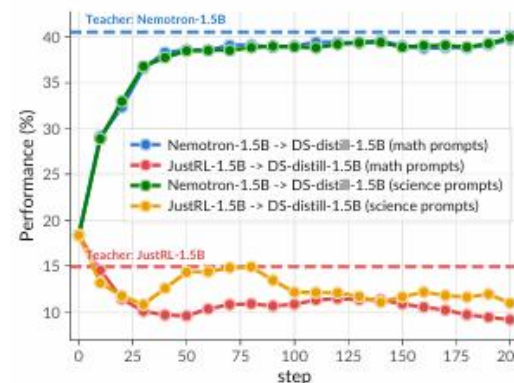
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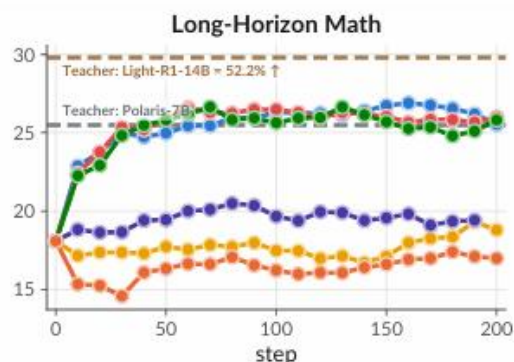
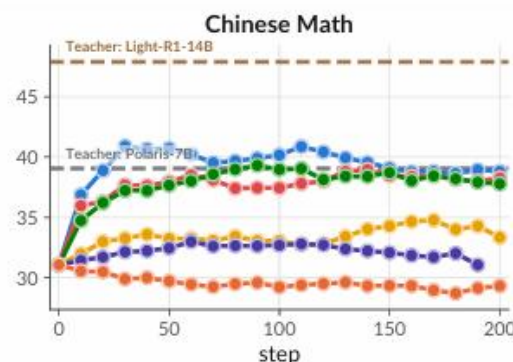
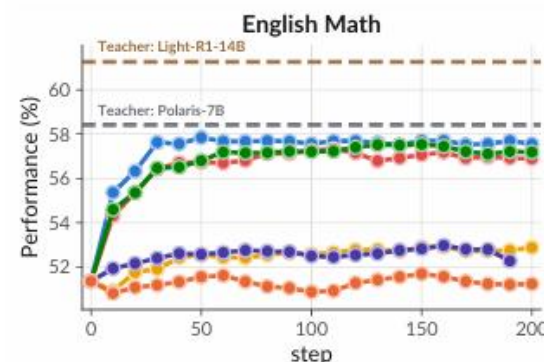
(a) LiveCodeBench (DS-distill-1.5B)



(b) LiveCodeBench (DS-distill-7B)



(c) GPQA-Diamond (DS-distill-1.5B)



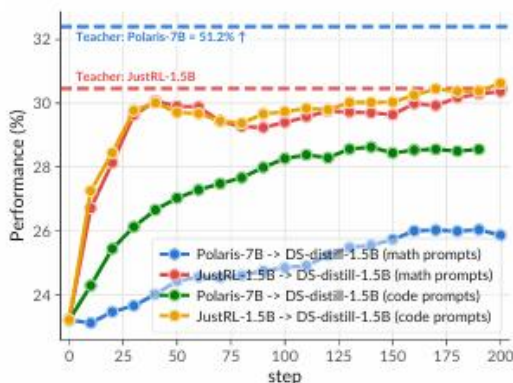
(d) Training on non-math domains, evaluating on math (DS-distill-7B)

只在math上训练, 能在完全没有 Code prompt 的情况下提高 code performance

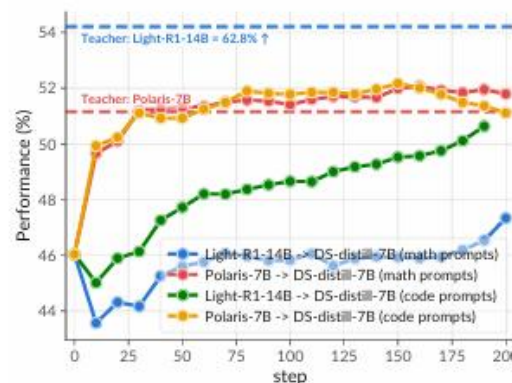
实验 3：领域泛化, math to code



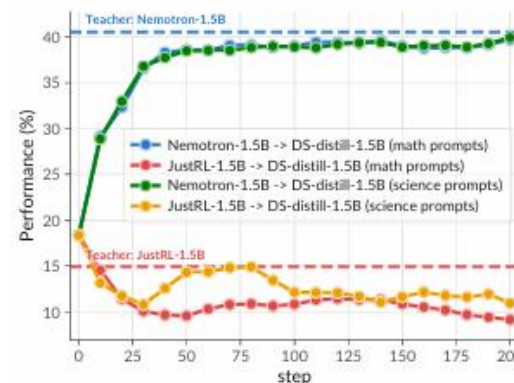
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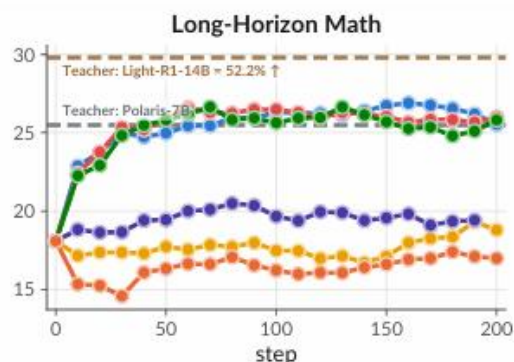
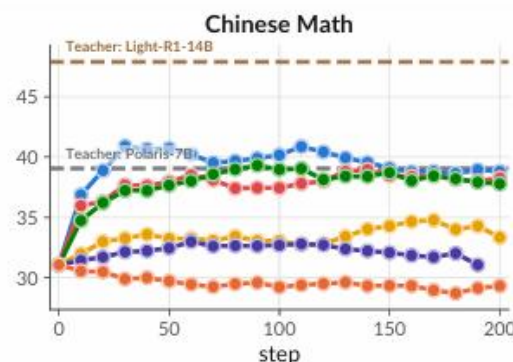
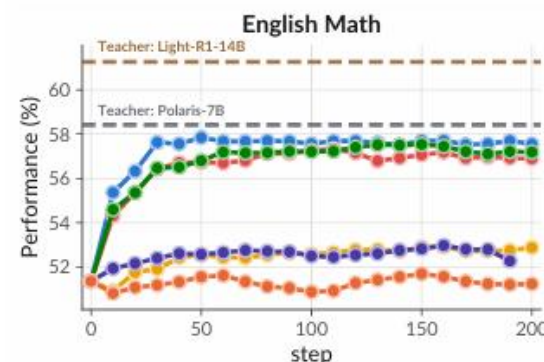
(a) LiveCodeBench (DS-distill-1.5B)



(b) LiveCodeBench (DS-distill-7B)



(c) GPQA-Diamond (DS-distill-1.5B)



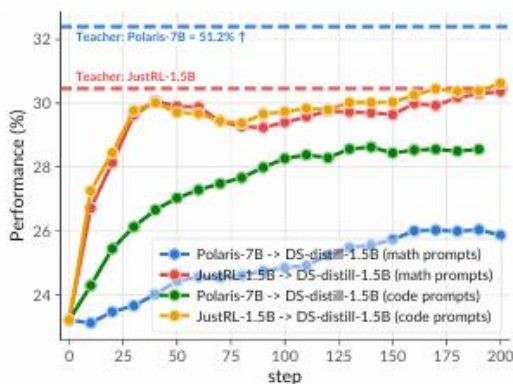
(d) Training on non-math domains, evaluating on math (DS-distill-7B)

对于 same-origin: 用 Math prompt OPD 和用 Code prompt OPD, LiveCodeBench 曲线相当接近
对于 cross-origin: Code prompt 训练明显优于 Math prompt 训练

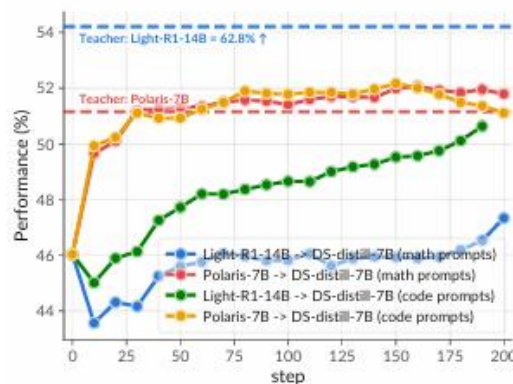
实验 3：领域泛化, math to code



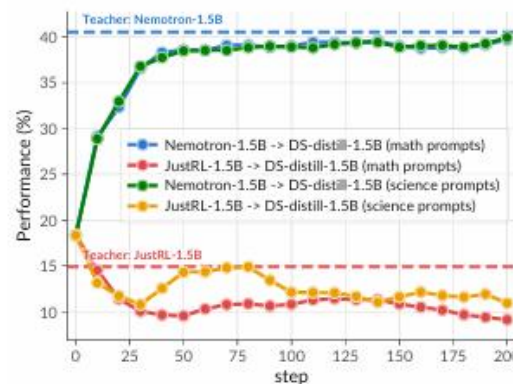
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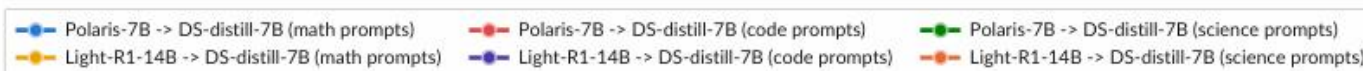
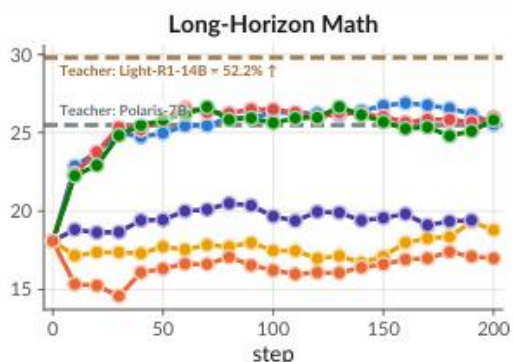
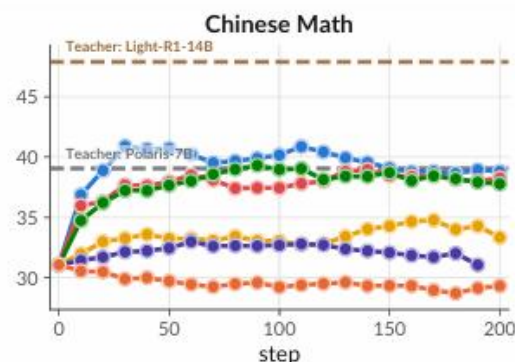
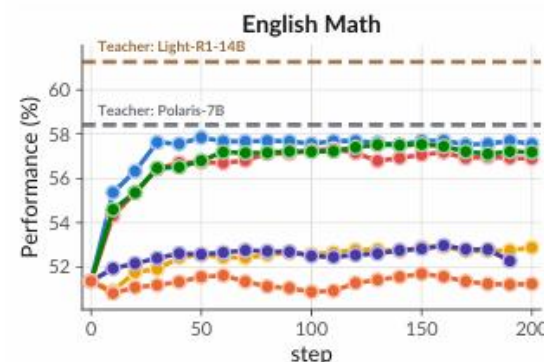
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(b) LiveCodeBench (DS-distill-7B)



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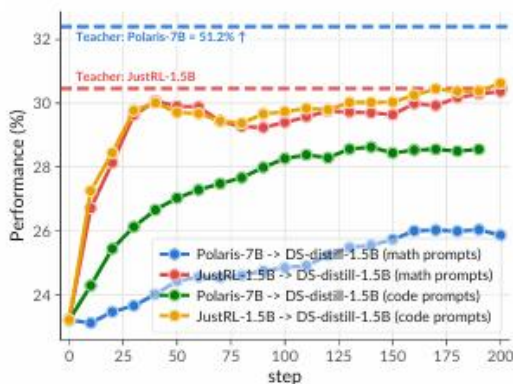
(ab)

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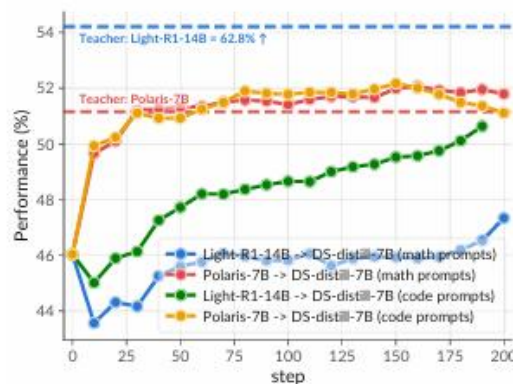
实验 3：领域泛化, math to science



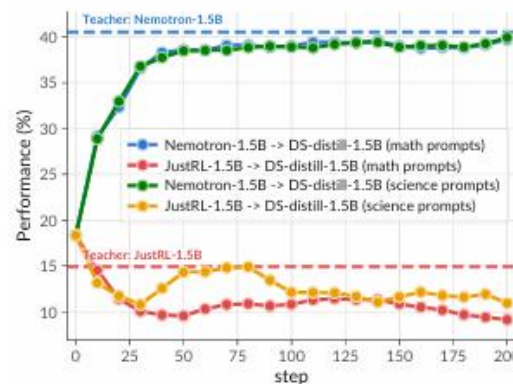
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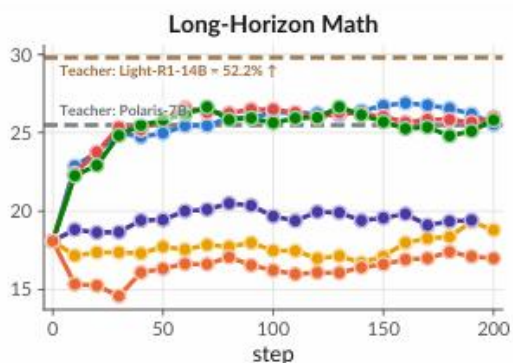
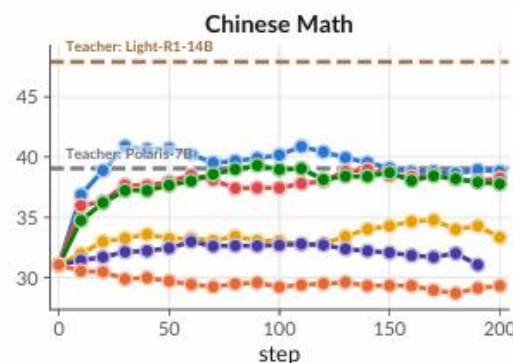
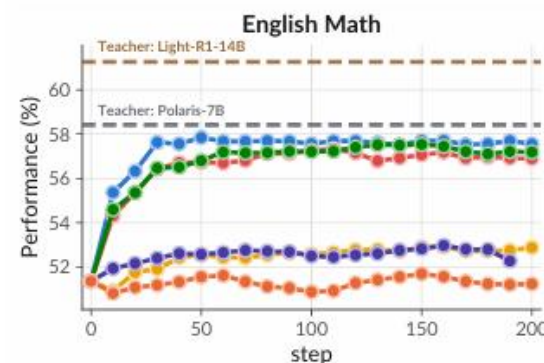
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(b) LiveCodeBench (DS-distill-7B)



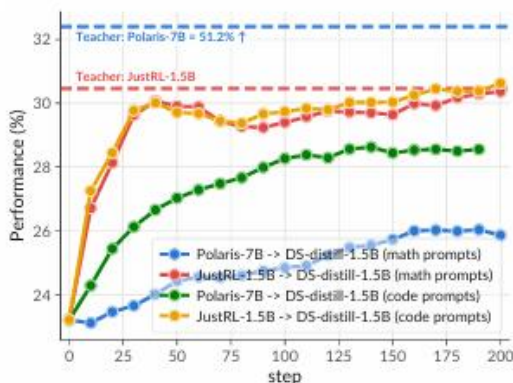
(c) GPQA-Diamond (DS-distill-1.5B)



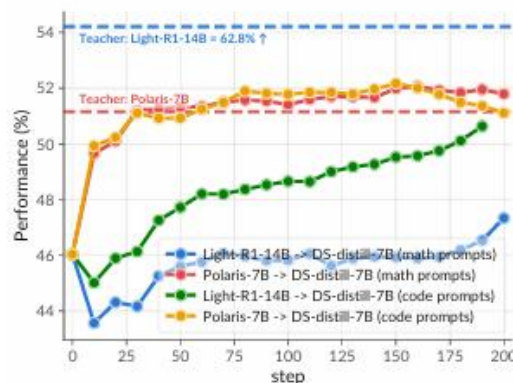
(d) Training on non-math domains, evaluating on math (DS-distill-7B)

(c) opd是一种 student capability→teacher capability
用math或者science都能把nemotron的能力蒸馏到student里
因为justrl的science本来更差, 所以不管用math还是用science, 都会让能力下降

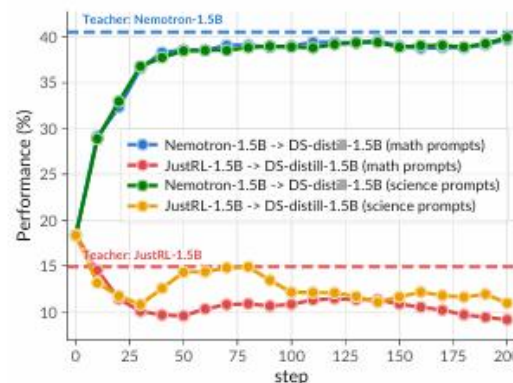
实验 3：领域泛化, math to science



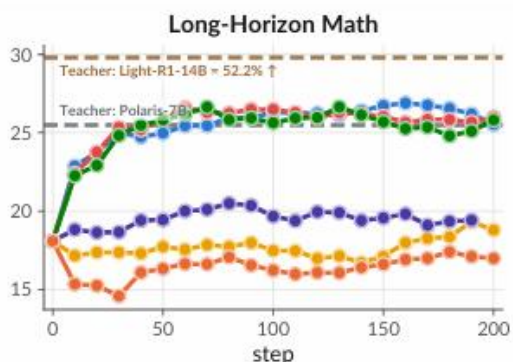
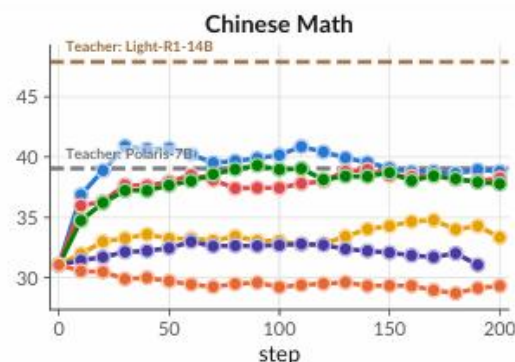
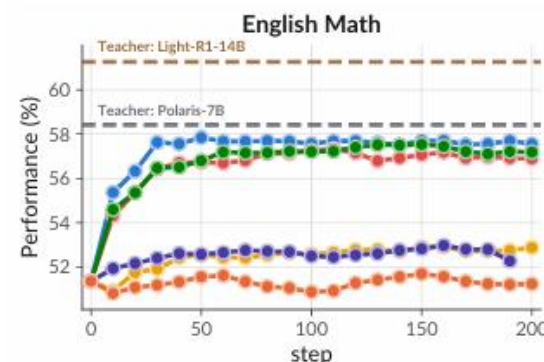
(a) LiveCodeBench (DS-distill-1.5B)



(b) LiveCodeBench (DS-distill-7B)



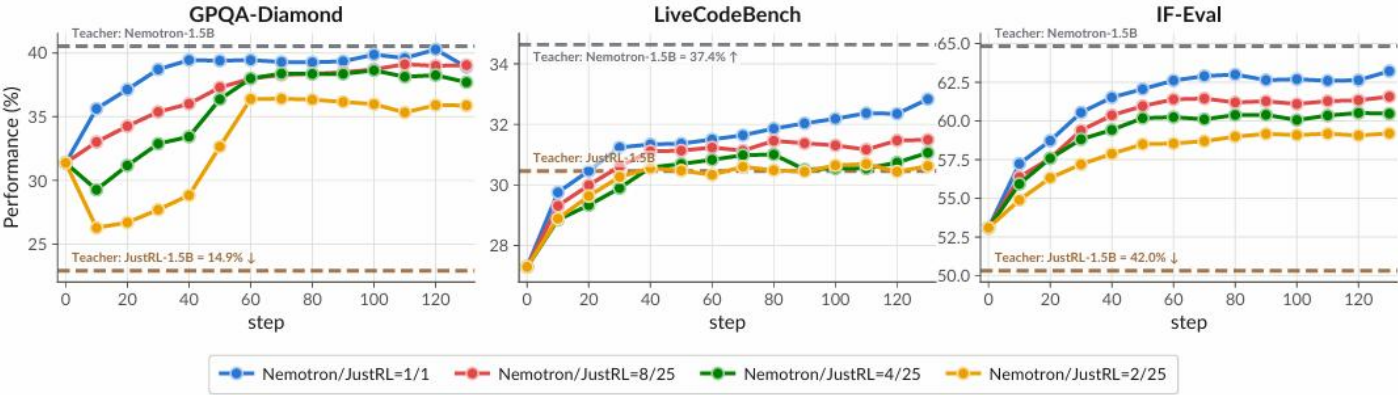
(c) GPQA-Diamond (DS-distill-1.5B)



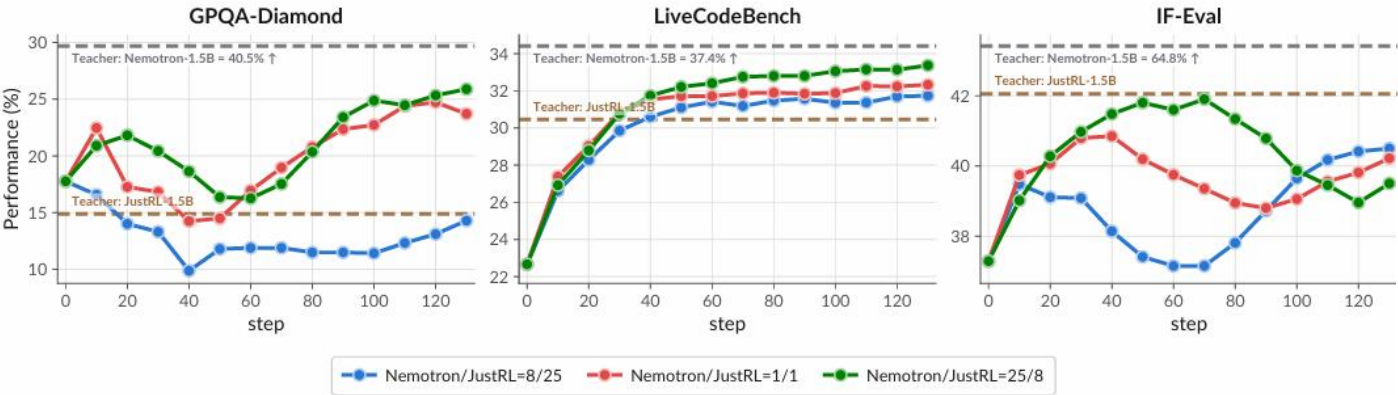
(d) Training on non-math domains, evaluating on math (DS-distill-7B)

(d) 反向同样成立：在code, science训练, math有提升, 甚至在IF上训练, math也有提升
同源提升可泛化, 不同源可能导致优化不充分/下降

实验4: MOPD (query 路由)



(a) Setting 1. Math teacher: JustRL-1.5B; Science/IF teacher: Nemotron-1.5B; student: Dev-1.5B.



(b) Setting 2. Math teacher: Nemotron-1.5B; Science/IF teacher: JustRL-1.5B; student: DS-distill-1.5B

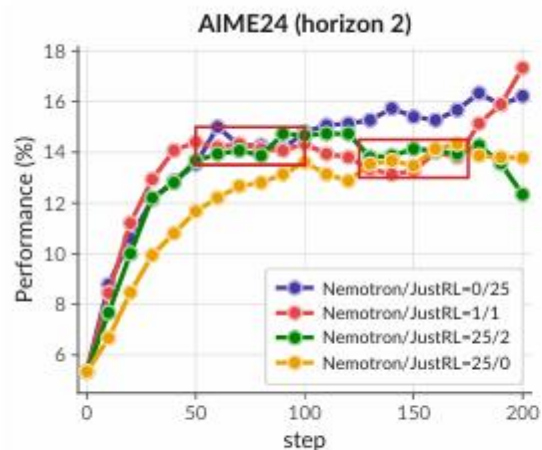
JustRL: math
Nemotron: science和if强

思想自由 兼容并包

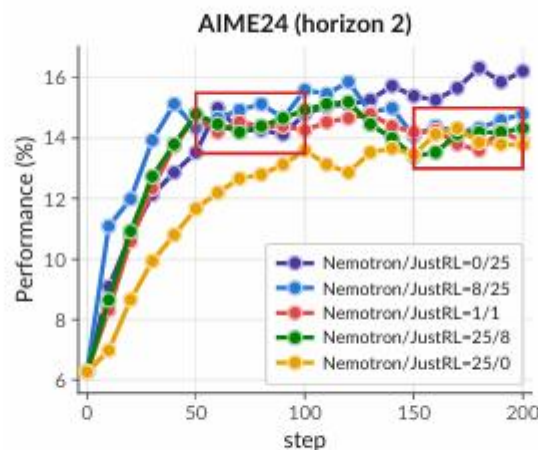
- (a) Math teacher JustRL 通过自己的 cross-domain influence 把 Science capability 拉低了
- (b) 交换teacher, 提升nemotron比例, science表现仍然朝nemotron靠近

Config	BeyondAIME	OlymMATH	Average
DS-distill-1.5B	9.6%	14.2%	11.9%(student)
Math Teacher: JustRL-1.5B, Science/IF Teacher: Nemotron-1.5B			
J/N = 25/2	19.4%	33.3%	26.4%(+0.1 pp)
J/N = 25/8	19.6%	34.5%	27.1%(+0.8 pp)
J/N = 1/1	20.2%	32.4%	26.3%(baseline)
J/N = 8/25	19.7%	32.5%	26.1%(-0.2 pp)
J/N = 2/25	19.2%	31.0%	25.1%(-1.2 pp)
Math Teacher: Nemotron-1.5B, Science/IF Teacher: JustRL-1.5B			
J/N = 25/8	17.4%	28.0%	22.7%(+0.7 pp)
J/N = 1/1	17.3%	26.6%	22.0%(baseline)
J/N = 8/25	17.1%	26.2%	21.7%(-0.3 pp)

实验4: MOPD (顺序)



(a) JustRL-Math, Nemotron-Science/IF



(b) JustRL-Science/IF, Nemotron-Math



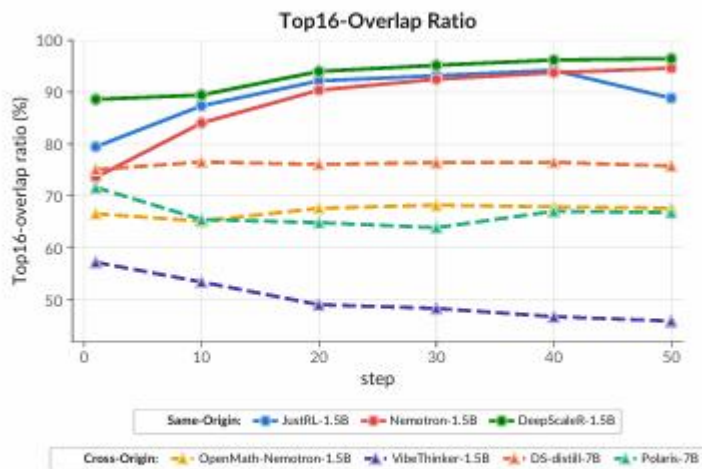
(c) Cascaded OPD, GPQA-Diamond

训练过程中可能会出现tug-of-war, 拉扯
MOPD把这种耦合在训练过程中体现了, 可以解耦成顺序MOPD观察这种现象

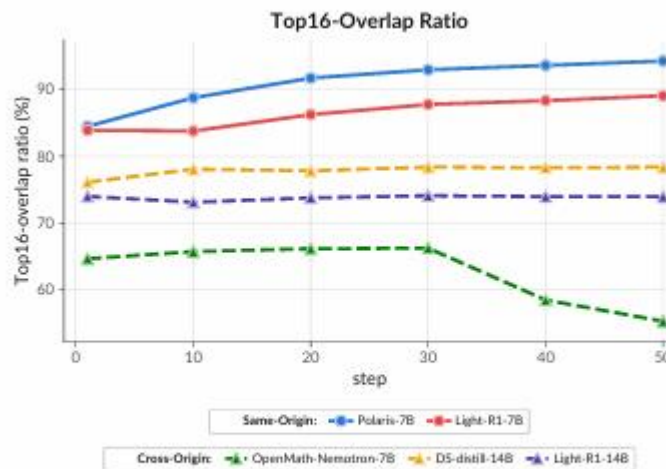
JustRL: math

Nemotron: science和if强

实验5: topk分析



(a) DS-distill-1.5B OPD Experiments



(b) DS-distill-7B OPD Experiments

训练前, same origin 的 top16就更高, 且随着训练进行更高不同origin的较难形成全局 policy agreement

实验6: origin改变杠杆长度



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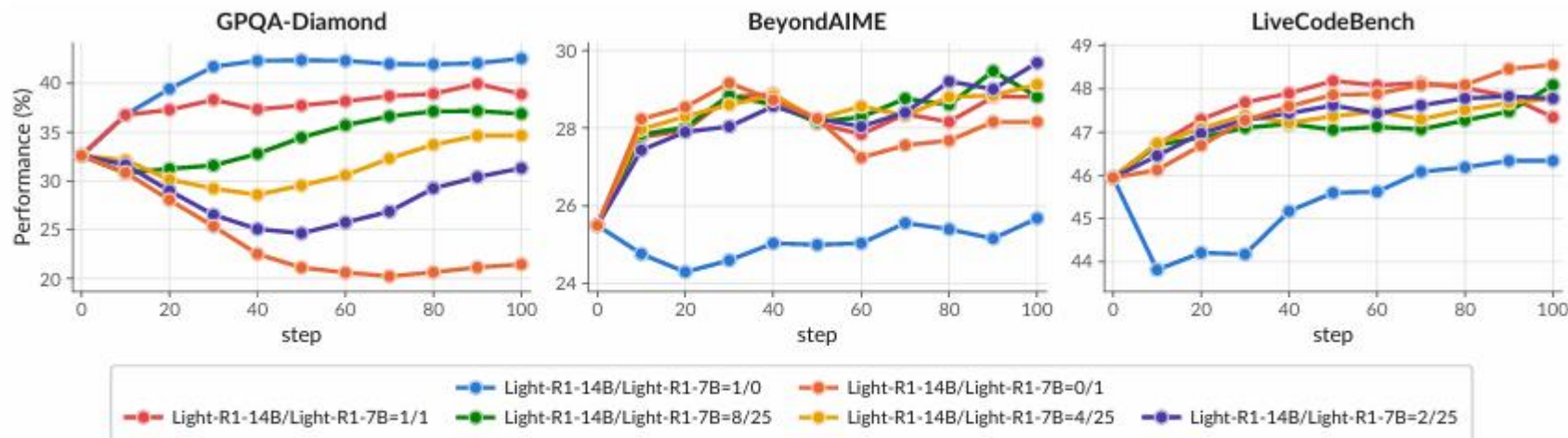


Figure 7. The MOPD experiment results of DS-distill-7B (student), Light-R1-7B (math teacher), and Light-R1-14B (science/IF teacher). The legends refer to the expert data mix ratio of different MOPD experiments ($\text{Light-R1-14B}/\text{Light-R1-7B} \in \{1/0, 1/1, 8/25, 4/25, 2/25, 0/1\}$).

即使在1:1的比例下，同源的“拉力”更强



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请批评指正